

CSCI 3052U: Machine Learning I

Group Project Instructions, Milestones, and Grading Rubric Fall 2026

1. Project Purpose and Learning Objectives

Overall Expectation

The group project is a semester-long, team-based machine-learning proof of concept. Each team will identify a meaningful real-world problem, obtain an appropriate dataset, build a reproducible machine-learning pipeline, compare models fairly, and communicate evidence-based conclusions.

The project is intended to develop both professional practice and research readiness. It is not necessary to produce publication-level novelty or state-of-the-art results. A carefully executed reproduction, comparative study, or applied project can earn full marks when the team demonstrates sound theory, rigorous experimentation, reproducible code, critical analysis, and clear communication.

A promising project may later be extended into independent research or a publication with the instructor's agreement; publication is an optional continuation and is not required for success in this course.

By completing the project, students will be able to:

- Formulate and pitch a feasible machine-learning problem by identifying the users or stakeholders, available data, constraints, risks, and measurable success criteria.
- Connect machine-learning theory to implementation by explaining the assumptions, objective functions, decision rules, optimization procedures, and limitations of the selected models.
- Implement an end-to-end machine-learning lifecycle in Python, including data preparation, exploratory analysis, preprocessing, model training, validation, evaluation, error analysis, and reporting.
- Establish and reproduce a defensible baseline before introducing more complex models or changes.
- Design controlled and fair experiments, select appropriate metrics, prevent data leakage, distinguish validation from test performance, and interpret uncertainty in results.
- Develop reusable and reviewable software using functions, classes, pipelines, version control, issue tracking, pair programming, and code review.
- Use generative-AI tools critically by verifying generated code and claims against theory, documentation, controlled experiments, and independently checked sources.
- Collaborate professionally, document individual contributions, respond to feedback, and present technical findings to both technical and non-technical audiences.

2. Team Size, Formation, and Shared Responsibility

Allowed team size: 3–5 students (to learn how to work in a team in industry and research). A different team size requires prior written approval from the instructor.

Team Contract

Each team must complete and submit a team contract with the first milestone. The contract must identify meeting expectations, communication channels, internal deadlines, decision-making procedures, expected weekly contributions, and the process for addressing missed work or conflict.

Teams may assign primary roles—such as project coordinator, data lead, modelling lead, evaluation/reproducibility lead, and communication lead—but roles must not become isolated silos. Every member is responsible for understanding the complete project and reviewing work that is incorporated into a group submission.

No student may claim that an error, unsupported statement, academic-integrity issue, or unverified AI-generated output is “someone else’s part.” All members must review the integrated submission before it is submitted.

3. Topic and Dataset Selection Rules

Project topics must be proposed by the students and approved by the instructional team. Projects may be motivated by an industry, community, research, or personally meaningful setting, provided that the problem is feasible within the semester and aligned with the methods taught in Machine Learning I.

Acceptable project forms include:

- A supervised classification or regression problem using tabular, text, image, time-series, or other approved data.
- An unsupervised-learning project involving clustering, dimensionality reduction, anomaly detection, or exploratory pattern discovery, with an approved evaluation plan.
- A reproduction of an existing baseline or published result, followed by a justified comparison, improvement, robustness check, or new analysis.
- A real-world proof of concept developed with an approved industry or community partner, provided that the team can meet all privacy, licensing, reproducibility, and assessment requirements.

The following are not sufficient on their own:

- A dashboard, visualization, or exploratory data analysis with no machine-learning question and no evaluated model.
- A copied Kaggle notebook, tutorial, or repository with only cosmetic changes.
- A no-code AutoML workflow, API wrapper, or prompt-only comparison in which the team cannot explain or evaluate the underlying method.
- A project that uses the test set repeatedly for model selection or reports only the best run without a documented selection procedure.
- A project that depends on inaccessible proprietary data, unapproved human-subject data collection, or computing resources that are not realistically available to the team.

Dataset approval requirements

- The dataset must have a lawful and documented source, license, or permission for course use.
- The team must not collect data from human participants or use identifiable, confidential, medical, employment, or other sensitive personal data without explicit instructor approval and any required ethics/privacy authorization. Such approval should not be assumed within the course timeline.
- The dataset must be large and rich enough to support meaningful evaluation, but small enough to process using the team’s approved resources.
- The team must document the unit of analysis, features, target or task, missingness, class balance or distribution, known limitations, and the proposed train/validation/test strategy.
- The same project or substantially the same work may not be submitted in another course unless all instructors provide prior written approval and the new contribution is clearly identified.

Lee Language Lab Project-List Bonus (5% bonus marks)

Eligibility and value. A team that selects an instructor-approved problem from the Lee Language Lab's current project list at Milestone 1 and receives final problem approval at Milestone 2 will receive a fixed bonus of five percentage points on the overall project grade.

Conditions. To remain eligible, the final submission must remain substantively aligned with the approved problem and satisfy all seven milestones and all ordinary technical, reproducibility, reporting, presentation, and individual-accountability requirements. Changing to an unrelated problem ends eligibility unless the instructor approves the change in writing.

Application and cap. The bonus is applied to each eligible student's final project grade after milestone grading and any individual accountability adjustment. The project grade is capped at 100%. Because the group project is worth 30% of the final course grade, the maximum effect of this bonus is 1.5 percentage points on the final course grade.

No implied research appointment. Selecting a Lee Language Lab problem does not guarantee lab membership, access to confidential data or dedicated computing resources, authorship, publication, paid work, or continued supervision. Any such opportunity requires a separate agreement.

4. Minimum Technical Requirements

Every project must include the following components unless the instructor approves an alternative in writing:

- A precise problem statement and a measurable primary success criterion established before final experimentation.
- A documented data pipeline covering acquisition, inspection, cleaning, preprocessing, feature construction or representation, and data splitting.
- A naive or task-specific benchmark and at least two substantive machine-learning models or modelling approaches.
- At least one model or method drawn directly from the course, such as k-nearest neighbours, linear or logistic regression, decision trees, ensembles, neural networks, Naive Bayes, clustering, Gaussian mixture models, or principal component analysis.
- A theoretical explanation of the selected methods. The report must explain at least one relevant mathematical or algorithmic element, such as a distance function, loss function, regularization term, split criterion, probabilistic assumption, gradient update, dimensionality-reduction objective, or evaluation equation.
- A fair model-selection and evaluation process that separates training, validation, and final testing and avoids leakage.
- Error analysis and discussion of limitations, including failure cases and possible sources of bias or unreliability.
- A reproducible repository and a final report and presentation that connect the evidence back to the original problem and success criterion.

Deep learning is optional rather than required. A team using a neural network or pretrained model must still include a simple classical baseline, justify the added complexity, document compute requirements, and demonstrate that each member understands the architecture and training procedure.

5. Required Baseline and Baseline Reproducibility

Starting-Out: Reproducible Baselines

Every team must establish a reproducible baseline before attempting a more complex model. The baseline is the reference point against which the value of later decisions is assessed.

The baseline submission must include:

- A naive benchmark appropriate to the task—for example, majority-class prediction, stratified random prediction, mean or median prediction, a one-cluster benchmark, or another justified reference.
- At least one implemented course-level baseline—for example, kNN, linear/logistic regression, a decision tree, Naive Bayes, k-means, or another approved method.
- The exact dataset version, split, preprocessing steps, random seeds, hyperparameters, software versions, and commands needed to reproduce the result.
- A table comparing the reproduced result with the expected or published result, where applicable, and a brief analysis of any discrepancy.
- A short demonstration in which any team member can run or explain the baseline pipeline.

A failed reproduction can still be valuable if the team documents what was attempted, isolates plausible causes, and provides evidence-based conclusions. Undocumented failure or an unverified copied result is not a reproducible baseline.

6. Required Model Comparison and Experimental Design

The final comparison must include, at minimum, one naive benchmark and two substantive machine-learning approaches. The models must be compared under the same data split, preprocessing assumptions, and evaluation protocol unless a difference is explicitly justified.

- Define the primary evaluation metric before final model selection. Additional metrics may be reported, but the team may not change the primary metric simply because another metric produces a more favourable ranking.
- Use a validation set, cross-validation, or another approved procedure for hyperparameter selection. The final test set should be used only for the final evaluation.
- Apply comparable tuning effort across models and report the search space or settings used.
- Use repeated runs, multiple random seeds, or cross-validation where results may be unstable. Report variability rather than only the single best run.
- Include at least one sensitivity, ablation, robustness, or subgroup analysis that answers a meaningful question about why the system behaves as observed.
- Record negative and inconclusive results. Do not delete unsuccessful runs or present exploratory test-set results as confirmatory evidence.

7. Evaluation Metrics and Critical Analysis

Teams must justify why their primary metric reflects the real objective and consequences of the task. Accuracy alone is not automatically appropriate, particularly for imbalanced data or unequal error costs.

Task Type	Examples of Appropriate Evidence
Classification	Accuracy where appropriate; macro/micro F1; precision; recall; ROC-AUC or PR-AUC when justified; confusion matrix; per-class or subgroup performance.

Task Type	Examples of Appropriate Evidence
Regression	MAE, RMSE, MSE, or R^2 as appropriate; residual analysis; comparison with mean/median prediction; practical interpretation of error magnitude.
Clustering / Unsupervised	Silhouette or another internal metric; ARI/NMI when reference labels are legitimately available; cluster-size and stability analysis; qualitative inspection of representative examples.
Recommendation / Ranking	Precision@k, Recall@k, MAP, MRR, or NDCG as appropriate; popularity baseline; cold-start or subgroup analysis where relevant.
All projects	Uncertainty or variability; computational cost; error/failure analysis; limitations; data leakage checks; and fairness, safety, or subgroup analysis when relevant to the application.

Results must be interpreted, not merely listed. Each major results paragraph should state: (1) the question or purpose, (2) the experimental approach, (3) the result, and (4) what the result does—and does not—support.

8. Reproducibility Requirements

A reader with access to the approved data should be able to reproduce the main reported result without private instructions from the team.

- README with project purpose, repository structure, setup instructions, data-access instructions, and exact commands or notebook sequence.
- requirements.txt, environment.yml, pyproject.toml, or another approved environment specification with package versions.
- Fixed and recorded random seeds, documented data splits, configuration files or clearly recorded hyperparameters, and an experiment log.
- A script or clearly ordered notebook workflow that regenerates the main results table or figure from the approved inputs.
- No hard-coded personal paths, passwords, tokens, credentials, or private data in the repository.
- A small reproducible example when the full dataset or computation cannot legally or practically be redistributed.
- Documentation of failed experiments and deviations from the approved plan when they affect interpretation.

9. GitHub or Repository Expectations

Each team must maintain a course Git repository or another instructor-approved version-control repository. The repository should be private by default and shared with the instructor and Teaching Assistants. A repository may be made public only when all team members consent and all data, code, model, license, privacy, and partner restrictions permit public release.

The repository must include:

- A clear directory structure separating source code, notebooks, configuration, documentation, tests or checks, reports, and outputs.
- Meaningful commit messages and a visible history of work throughout the term; a single bulk upload near the deadline is not sufficient evidence of collaboration.
- Issues, a project board, or another task tracker showing owners, deadlines, blockers, and decisions.

- Branching and pull-request practice for substantive changes. Each student must submit at least one meaningful pull request and complete at least one constructive code review of another member's work.
- A LICENSE or clear reuse statement for original code and appropriate attribution for external code, data, and models.
- A CONTRIBUTIONS.md file or equivalent contribution statement identifying each member's work.
- An AI_USE.md file or report appendix documenting permitted generative-AI assistance.

10. Rules for Using Generative AI

Generative AI may assist the project only where permitted by the course and milestone instructions. It is a fallible tool, not an authoritative source or substitute for understanding.

Permitted uses may include:

- Brainstorming project alternatives, debugging strategies, test cases, or questions to investigate.
- Explaining an error message or unfamiliar code, provided that the team verifies the explanation against documentation and experiments.
- Suggesting code structure, comments, unit tests, or editing improvements that the team independently reviews and understands.
- Practising presentation questions or generating a checklist for review.

Prohibited or unacceptable uses include:

- Submitting AI-generated code, prose, figures, analyses, or answers that the team cannot independently explain and defend.
- Using generative AI to invent, select, or summarize citations without personally locating, reading, and verifying the sources.
- Reporting generated numbers, experiments, outputs, or claims that were not actually produced and checked by the team.
- Uploading confidential, copyrighted, personally identifiable, partner-restricted, or otherwise sensitive data to an external AI service.
- Using AI to conceal unequal contribution, fabricate progress, or bypass an individual assessment or oral defence.

The final submission must include a concise AI-use statement naming the tool, purpose, material used, and verification performed. "No generative AI used" is an acceptable statement. Undisclosed or unverified use may be addressed under the academic-integrity policy.

11. Individual Accountability and Contribution Evidence

The project is collaborative, but assessment includes individual accountability. Group membership does not guarantee identical grades.

- Each student must contribute to both technical work and the review or communication of the project; one student may not complete all modelling while another contributes only formatting.
- Teams must maintain meeting notes, task assignments, issue history, pull requests, code reviews, and a final contribution statement.

- Peer contribution check-ins will occur at the beginning, midpoint, and end of the project. Peer reports are evidence for instructor review rather than an automatic popularity-based grade multiplier.
- Any student may be asked to explain code, theory, data decisions, experimental design, results, limitations, or AI-assisted work from any part of the submission.
- The individual reflection and oral defence are completed independently. A student who cannot demonstrate understanding of the submitted work may receive a lower individual project grade even when the team artifact is strong.

12. Milestones, Deliverables, and Proposed Due Dates

The dates below are aligned with the current Fall 2026 course calendar and are proposed for the final syllabus. Canvas is the official source for confirmed dates, submission links, formats, and any approved changes.

#	Milestone	Course (Weight)	Proposed Due Date	Required Evidence	Submission
1	Proposal and Team Contract	3%(10%)	Fri. Sept. 18	2–3 page proposal; team contract; initial roles; task, current practice, gap, proposed solution, success evidence; SMART timeline.	One PDF + contract + repository link through Canvas.
2	Dataset and Problem Approval	3%(10%)	Fri. Oct. 2	Data source/license; data card; sample inputs/outputs; initial EDA; target/task; risks; train/validation/test strategy; primary metric; feasibility and compute plan.	Approval memo/notebook + updated proposal and repository.
3	Baseline Reproduction	3%(10%)	Mon. Oct. 19	Working data pipeline; naive benchmark; course-level baseline; reproducible environment; baseline table; discrepancy analysis; start/stop/continue plan.	Repository tag/release + maximum 2-page technical memo + lab demonstration.
4	Progress Presentation	3%(10%)	Fri. Oct 30	Current methods; theoretical justification; preliminary results; deviations from plan; blockers; revised timeline; response to feedback; midpoint peer check-in.r	Slides/video (5-minute recording) + slides + progress repot.

#	Milestone	Course (Weight)	Proposed Due Date	Required Evidence	Submission
5	Final Experiments and Analysis	6%(20%)	Fri. Nov 20	Fair comparison of required models; tuning procedure; uncertainty; main table/figure; error analysis; sensitivity/r obustness/ablation; draft conclusions and limitations.	Results package + experiment log + draft Results section.
6	Final Report, Repository, and Demonstration	9%(30%)	Fri. Dec 4	8–10 minute presentation/demo; 6–8 page final report; reproducible repository; main result regeneration; limitations; contribution statement; AI-use statement; response to feedback.	Slides/video (5-minute recording) + PDF report + repository release + 1-page revision response.
7	Individual Reflection and Oral Defence	3%(10%)	Mon. Dec. 7	1–2 page individual reflection; contribution evidence; key theoretical and experimental decisions; one failure/lesson; responsible AI use; 5 minute oral or code walkthrough.	Individual submission; no collaboration on reflection or defence. Oral video/slide walkthrough

Base total: 100% of the project grade = 30% of the final course grade. Eligible Lee Language Lab project-list teams receive +5 project percentage points after individual adjustments, capped at a final project grade of 100% (maximum effect: +1.5 percentage points on the final course grade) (NOT AN EIGHTH MILESTONE)

Eligibility is declared in Milestone 1, confirmed in Milestone 2, and verified again at final grading. The seven milestone weights remain the complete 30% base project assessment. The bonus is capped so that the final project grade cannot exceed 100%.

13. Final Report Requirements

Project Milestone 6. Final Presentation and Final Report Writeup

The final report should be 6–8 pages, excluding references and appendices, using the course-provided scientific-report template. Appendices may contain supplementary experiments, derivations, code excerpts, questionnaires, or additional figures, but the main argument and required evidence must appear in the main report.

Required sections:

- Abstract: problem, approach, data, and one or two main findings.
- Introduction and motivation: intended setting, why the problem matters, task, contribution, and success criterion.
- Related work or technical context: approximately 5–8 verified and relevant scholarly or authoritative sources; clearly distinguish prior work from the team's work.

- Data and task: inputs/outputs, data source and license, data statistics, preprocessing, split procedure, risks, and limitations.
- Methods and theory: baselines, comparison models, key assumptions, equations or algorithmic description, and justification of design choices.
- Experimental setup: environment, hyperparameters, tuning procedure, seeds/folds, metrics, and reproducibility details.
- Results and analysis: main tables/figures, uncertainty, comparison to baselines, error analysis, sensitivity/robustness evidence, and interpretation.
- Responsible-use and limitations statement: bias, privacy, fairness, reliability, failure cases, compute, scope, and appropriate/non-appropriate use.
- Conclusion: what the evidence supports, what remains uncertain, and feasible next steps.
- Contribution and AI-use statements.

The report is graded on the quality of the reasoning and evidence rather than whether the most complex model wins. A negative or inconclusive result can receive a strong grade when the experiment is well designed, reproducible, and critically analyzed.

14. Presentation and Demonstration Requirements

Course Presentation to Showcase the Project

- The final presentation should be 8–10 minutes, followed by questions. All members must speak and may be questioned about any part of the work.
- Present the problem, intended users, data, baseline, model comparison, main evidence, limitations, and one practical or research takeaway.
- Demonstrate the working pipeline or a stable recorded walkthrough. Teams should prepare a backup that does not depend on live internet access or an external service.
- Use figures and tables that make the main result understandable to a third-year ML audience and an invited industry or research panel.
- Submit slides and any required video backup by the deadline posted on Canvas. Feedback from the presentation must be addressed in the final report and revision-response document.

15. Peer Contribution and Final Contribution Statement

Each final report must include a contribution statement using clear activity categories, such as problem formulation, data preparation, software, modelling, experiments, analysis, visualization, project management, writing, and presentation.

Before submission, every member must confirm that the contribution statement is accurate and that they have reviewed the full submission. Disagreement must be reported to the instructor before the deadline; the team may not unilaterally remove a member or alter a member's grade.

Peer evaluations, repository evidence, meeting records, milestone participation, and oral questioning may be used together to determine individual adjustments. No single peer score will be used as the sole basis for an individual grade.

16. Late Work and Team-Conflict Procedures

Late work is governed by the course syllabus and any approved accommodation. Group membership does not create a separate extension policy, and one member's delay does not automatically extend the deadline for the full team.

For team difficulties, use the following escalation process:

- 1 Discuss the issue promptly and refer to the team contract, task tracker, and agreed internal deadlines.
- 2 Record the concern, the expected corrective action, an owner, and a short deadline in the team's project record.
- 3 If the concern is not resolved within three days—or if two important commitments are missed—contact the Teaching Assistant with a concise factual summary and supporting records. Do not wait until the final week.
- 4 Participate in any required remediation meeting. The instructor may revise responsibilities, establish a recovery plan, require additional individual evidence, split work, adjust individual grades, or, in serious cases, remove a student from the group project.
- 5 Report harassment, discrimination, threats, privacy breaches, fabrication, plagiarism, or other academic-integrity concerns immediately rather than using the ordinary conflict timeline.

Teams may not conceal missing work, fabricate contribution records, submit another member's work without review, or retaliate against a student who raises a legitimate concern.

17. Overall Grading Rubric

The rubric below applies across the seven milestones. Early milestones assess planning and readiness; later milestones assess execution, evidence, communication, and individual understanding. The seven milestones together contribute 30% to the final course grade. The Lee Language Lab project-list bonus is applied separately under the eligibility rules above.

Criterion	Full-Credit Standard	Common Reasons for Deduction
Problem formulation and value	Problem is clear, feasible, meaningful, scoped to the semester, and connected to measurable success criteria and intended users.	Vague task; no stakeholder or success criterion; scope is trivial or infeasible; claims are disconnected from the actual ML task.
Data quality, legality, and ethics	Data source, license, unit of analysis, statistics, split, limitations, privacy, and risks are documented and handled appropriately.	Undocumented or inappropriate data; unclear target; leakage; unapproved sensitive data; missing license/permission; weak data understanding.
Theory and method justification	Models are explained accurately using relevant assumptions, equations, objectives, or algorithms; choices are justified from theory and task requirements.	Library calls without understanding; incorrect theory; unsupported model choice; unnecessary complexity; cannot explain how the method works.
Baseline and experimental design	Naive and course-level baselines are reproducible; comparisons are fair; tuning and validation are documented; test set is protected.	No defensible baseline; unfair comparison; hidden tuning; repeated test-set use; cherry-picked runs; unclear experimental controls.
Implementation and software quality	Pipeline is correct, modular, readable, reviewable, and robust; code uses appropriate abstractions, checks, version control, and documentation.	Broken or copied code; hard-coded paths; no reusable structure; minimal commit history; missing tests/checks; code cannot be explained.

Criterion	Full-Credit Standard	Common Reasons for Deduction
Evaluation, results, and critical analysis	Metrics are justified; variability and main evidence are clear; results are interpreted; errors, limitations, robustness, and negative findings are analyzed honestly.	Metrics do not match the problem; only best score reported; no uncertainty or error analysis; conclusions exceed the evidence; results merely listed.
Reproducibility and repository	Environment, data instructions, seeds, configs, commands, logs, main-result regeneration, contribution record, and AI-use record are complete.	Main result cannot be reproduced; missing dependencies/data instructions; secrets or restricted data exposed; repository assembled only at the end.
Report, presentation, and response to feedback	Writing and visuals are concise and technically accurate; presentation is accessible; feedback is addressed transparently and specifically.	Disorganized report; unreadable figures; unsupported claims; uneven presentation; no meaningful response to feedback.
Teamwork and individual understanding	Work is distributed and reviewed; communication is professional; contribution evidence is credible; each student can independently defend the project.	Persistent non-participation; isolated role silos; inaccurate contribution claims; inability to explain submitted work; unresolved conduct concerns.